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Online Videos Week 5

Topic 2: Torsion, Bending & Beam Elements

KB5034 Mechanics and Finite Element Analysis

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Learning Outcome 2.3

By the end of this lesson, students will be able to…

Describe the differences between beam elements and truss elements in terms of the number of nodes, degrees of freedom and the size of the stiffness matrix.

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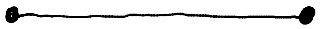
2D Beam Element

ϕ =

nodes

DOF at each node

Counter clockwise is opposite to traditional beam theory



$$\begin{bmatrix} \quad \\ \quad \\ \quad \\ \quad \\ \quad \\ \quad \end{bmatrix} = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} \quad \\ \quad \\ \quad \\ \quad \\ \quad \\ \quad \end{bmatrix}$$

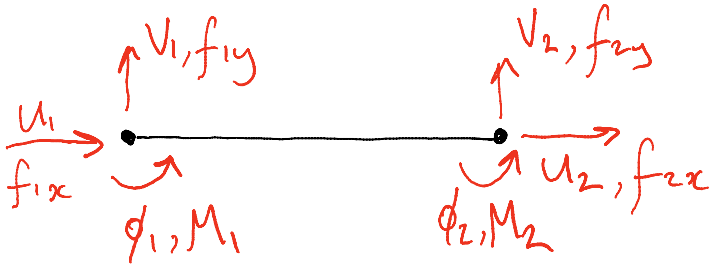
Total DOF =

Therefore,

Size of $\underline{k}_e$ =

x

2D Beam Elements



for a truss element:

$$\begin{bmatrix} f_{1x} \\ f_{2x} \end{bmatrix} = \begin{bmatrix} \frac{EA}{\ell} & -\frac{EA}{\ell} \\ -\frac{EA}{\ell} & \frac{EA}{\ell} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} f_{1x} \\ f_{1y} \\ M_1 \\ f_{2x} \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} & & & & & \\ & k_{22} & k_{23} & & k_{25} & k_{26} \\ & k_{32} & k_{33} & & k_{35} & k_{36} \\ & & & & & \\ & k_{52} & k_{53} & & k_{55} & k_{56} \\ & k_{62} & k_{63} & & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} u_1 \\ v_1 \\ \phi_1 \\ u_2 \\ v_2 \\ \phi_2 \end{bmatrix}$$

Superposition i.e

$$\begin{bmatrix} f_{1y} \\ M_1 \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} k_{22} & k_{23} & k_{25} & k_{26} \\ k_{32} & k_{33} & k_{35} & k_{36} \\ k_{52} & k_{53} & k_{55} & k_{56} \\ k_{62} & k_{63} & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} v_1 \\ \phi_1 \\ v_2 \\ \phi_2 \end{bmatrix}$$

Unknown k_{ij} terms

Unit Displacement Method

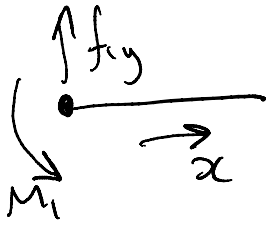
Step 1 $v_1 = 1, \ \phi_1 = v_2 = \phi_2 = 0$

$$\begin{bmatrix} f_{1y} \\ M_1 \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} k_{22} & k_{23} & k_{25} & k_{26} \\ k_{32} & k_{33} & k_{35} & k_{36} \\ k_{52} & k_{53} & k_{55} & k_{56} \\ k_{62} & k_{63} & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} & \\ & \\ & \\ & \end{bmatrix} \longrightarrow \begin{bmatrix} f_{1y} \\ M_1 \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} & \\ & \\ & \\ & \end{bmatrix}$$

Use beam theory to solve

Euler-Bernoulli Beam Theory

$M =$



traditional beam theory

↻ = +ve

Integrate to get slope

At $x=0 \rightarrow \quad v(0) = \qquad , \quad \frac{dv}{dx}(0) =$

Integrate to get deflection

Two equations

①

②

At node 2, $x = l \rightarrow \quad v(l) = \qquad , \quad \frac{dv}{dx}(l) =$

③

④

Rearrange ③

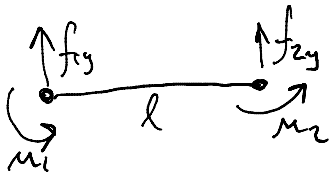
$$M_1 =$$

Sub into ④

$$\begin{bmatrix} f_{1y} \\ M_1 \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} k_{22} \\ k_{52} \\ k_{52} \\ k_{62} \end{bmatrix}$$

from UDM

Equilibrium at node 2:



So far we have calculated:

$$\begin{bmatrix} f_{1x} \\ f_{1y} \\ M_1 \\ f_{2x} \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} \frac{EA}{\ell} & 0 & 0 & -\frac{EA}{\ell} & 0 & 0 \\ 0 & & k_{33} & & k_{35} & k_{36} \\ 0 & & k_{53} & & k_{55} & k_{56} \\ -\frac{EA}{\ell} & 0 & 0 & \frac{EA}{\ell} & 0 & 0 \\ 0 & & k_{53} & 0 & k_{55} & k_{56} \\ 0 & & k_{63} & 0 & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} u_1 \\ v_1 \\ \phi_1 \\ u_2 \\ v_2 \\ \phi_2 \end{bmatrix}$$

$$\begin{bmatrix} f_{2y} \\ M_2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \begin{matrix} k_{53} & 0 & k_{55} & k_{56} \\ k_{63} & 0 & k_{65} & k_{66} \end{matrix} \begin{bmatrix} v_2 \\ \phi_2 \end{bmatrix}$$

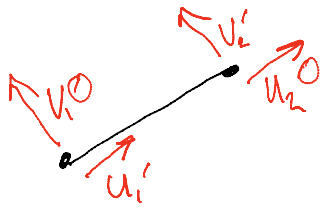
step 2 $v_1 = 0, \quad \phi_1 = 1, \quad v_2 = \phi_2 = 0$

Step 3 $v_1 = \phi_1 = 0, \quad v_2 = 1, \quad \phi_2 = 0$

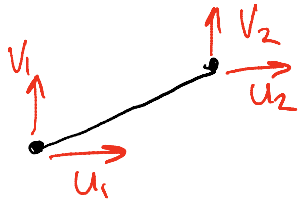
Step 4 $v_1 = \phi_1 = v_2 = 0, \quad \phi_2 = 1$

$$\begin{bmatrix} f_{1x} \\ f_{1y} \\ M_1 \\ f_{2x} \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} \frac{EA}{\ell} & 0 & 0 & -\frac{EA}{\ell} & 0 & 0 \\ 0 & \frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} & 0 & -\frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} \\ 0 & \frac{6EI}{\ell^2} & \frac{4EI}{\ell} & 0 & -\frac{6EI}{\ell^2} & \frac{2EI}{\ell} \\ -\frac{EA}{\ell} & 0 & 0 & \frac{EA}{\ell} & 0 & 0 \\ 0 & -\frac{12EI}{\ell^3} & -\frac{6EI}{\ell^2} & 0 & \frac{12EI}{\ell^3} & -\frac{6EI}{\ell^2} \\ 0 & \frac{6EI}{\ell^2} & \frac{2EI}{\ell} & 0 & -\frac{6EI}{\ell^2} & \frac{4EI}{\ell} \end{bmatrix} \begin{bmatrix} u_1 \\ v_1 \\ \phi_1 \\ u_2 \\ v_2 \\ \phi_2 \end{bmatrix}$$

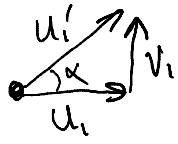
Axial deformation (u_1, u_2) & bending are decoupled







Transformation Matrix Beam Elements



$u_1' =$



$v_1' =$

$\phi_1' =$

$$u_2' =$$

$$v_2' =$$

$$\phi_2' =$$

In Matrix form:

$$\begin{bmatrix} u_1' \\ v_1' \\ \phi_1' \\ u_2' \\ v_2' \\ \phi_2' \end{bmatrix} = \begin{bmatrix} & & \\ & T & \\ & & \end{bmatrix} \begin{bmatrix} u_1 \\ v_1 \\ \phi_1 \\ u_2 \\ v_2 \\ \phi_2 \end{bmatrix}$$

$$T^{-1} = T^T$$

$$\underline{f_e} = \underline{T}^T \underline{k_e'} \underline{T} \underline{u_e}$$

Stiffness in global

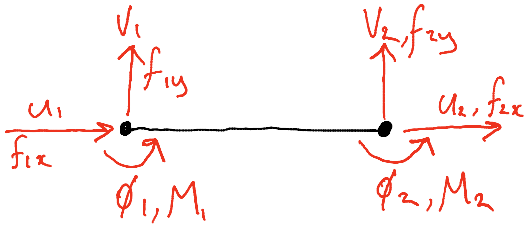
Learning Outcome 2.5

By the end of this lesson, students will be able to…

Derive the stiffness matrix of a beam element by the unit displacement method and by the method of Virtual Work.

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2D Beam Element



for a truss element:

$$\begin{bmatrix} f_{1x} \\ f_{2x} \end{bmatrix} = \begin{bmatrix} \frac{EA}{L} & -\frac{EA}{L} \\ -\frac{EA}{L} & \frac{EA}{L} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{bmatrix} f_{1x} \\ f_{1y} \\ M_1 \\ f_{2x} \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} & & & & & \\ & k_{22} & k_{23} & k_{25} & k_{26} \\ & k_{32} & k_{33} & k_{35} & k_{36} \\ & & & & & \\ & k_{52} & k_{53} & k_{55} & k_{56} \\ & k_{62} & k_{63} & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} u_1 \\ v_1 \\ \phi_1 \\ u_2 \\ v_2 \\ \phi_2 \end{bmatrix}$$

Superposition i.e axial deformation does not cause transverse deformation or rotation.

$$\begin{bmatrix} \\ \\ \\ \end{bmatrix} = \begin{bmatrix} k_{22} & k_{23} & k_{25} & k_{26} \\ k_{32} & k_{33} & k_{35} & k_{36} \\ k_{52} & k_{53} & k_{55} & k_{56} \\ k_{62} & k_{63} & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} \\ \\ \\ \end{bmatrix}$$

} Unknown k_{ij} terms

Unit Displacement Method

Step 1 $v_1 = 1, \quad \phi_1 = v_2 = \phi_2 = 0$

$$\begin{bmatrix} f_{1y} \\ M_1 \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} k_{22} & k_{23} & k_{25} & k_{26} \\ k_{32} & k_{33} & k_{35} & k_{36} \\ k_{52} & k_{53} & k_{55} & k_{56} \\ k_{62} & k_{63} & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} \\ \\ \\ \end{bmatrix} \rightarrow$$

Use beam theory to solve.

Euler-Bernoulli Beam Theory

$M =$



traditional beam theory
↻ = +ve

Integrate to get slope

Unit displacement method

$V(x=0) =$, $\frac{dV}{dx}(x=0) =$

At $x = 0 \rightarrow$

Integrate to get deflection

At $x = 0 \rightarrow$

Two equations:

①

②

At $x = l \rightarrow V(l) =$, $\phi(l) =$ (UDM)

from ①

from ②

} 2 equations, 2 unknowns (f_{iy}, M_i)

Rearrange ①

$M_1 =$

Sub into ②

$f_{1y} =$

Sub into ①

$M_1 =$

Equilibrium of force and moments at node 2:

Therefore:

$f_{2y} =$

$M_2 =$

$M_2 =$

from unit displacement method: $\begin{bmatrix} f_{1y} \\ M_1 \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} k_{22} \\ k_{32} \\ k_{52} \\ k_{62} \end{bmatrix}$

$k_{22} =$

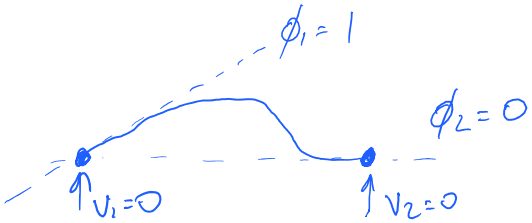
$k_{32} =$

$k_{52} =$

$k_{62} =$

Step 2

$V_1 = 0, \ \phi_1 = 1, \ V_2 = \phi_2 = 0$



from $\underline{f_e} = \underline{k_e u_e}$:

$$\begin{bmatrix} \quad \\ \quad \\ \quad \\ \quad \end{bmatrix} = \begin{bmatrix} f_{1y} \\ M_1 \\ f_{2y} \\ M_2 \end{bmatrix}$$

Beam theory:

$$EI\frac{d^2v}{dx^2} = -M_1 + f_{1y}x$$

Integrate to get slope:

Integrate to get deflection:

Boundary conditions:

at $x=0$: $V(0)=$, $\phi(0)=$

Two equations:

At $x=l$: $V(l)=$, $\phi(l)=$

2 equations

2 unknowns

Rearrange and solve:

$f_{1y}=$, $M_1=$

Equilibrium at node 2:

$f_{1y}=-f_{2y}$

$M_1+M_2-f_{1y}l=0$

$f_{2y}=$, $M_2=$

From unit displacement:

$$\begin{bmatrix} k_{23} \\ k_{33} \\ k_{53} \\ k_{63} \end{bmatrix} = \begin{bmatrix} f_{1y} \\ M_1 \\ f_{2y} \\ M_2 \end{bmatrix}$$

$k_{23}=$ $k_{33}=$

$k_{53}=$ $k_{63}=$

So far we have calculated:

$$\begin{bmatrix} f_{1x} \\ f_{1y} \\ M_1 \\ f_{2x} \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} \frac{EA}{\ell} & 0 & 0 & -\frac{EA}{\ell} & 0 & 0 \\ 0 & k_{22} & k_{23} & 0 & k_{25} & k_{26} \\ 0 & k_{32} & k_{33} & 0 & k_{35} & k_{36} \\ -\frac{EA}{\ell} & 0 & 0 & \frac{EA}{\ell} & 0 & 0 \\ 0 & k_{52} & k_{53} & 0 & k_{55} & k_{56} \\ 0 & k_{62} & k_{63} & 0 & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} u_1 \\ v_1 \\ \phi_1 \\ u_2 \\ v_2 \\ \phi_2 \end{bmatrix}$$

$$\begin{bmatrix} f_{1x} \\ f_{1y} \\ M_1 \\ f_{2x} \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} \frac{EA}{\ell} & 0 & 0 & -\frac{EA}{\ell} & 0 & 0 \\ 0 & & & 0 & k_{25} & k_{26} \\ 0 & & & 0 & k_{35} & k_{36} \\ -\frac{EA}{\ell} & 0 & 0 & \frac{EA}{\ell} & 0 & 0 \\ 0 & & & 0 & k_{55} & k_{56} \\ 0 & & & 0 & k_{65} & k_{66} \end{bmatrix} \begin{bmatrix} u_1 \\ v_1 \\ \phi_1 \\ u_2 \\ v_2 \\ \phi_2 \end{bmatrix}$$

Use other steps in UDM to calculate remaining terms.
($v_2 = 1$ & $\phi_2 = 1$)

$$\begin{bmatrix} f_{1x} \\ f_{1y} \\ M_1 \\ f_{2x} \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} \frac{EA}{\ell} & 0 & 0 & -\frac{EA}{\ell} & 0 & 0 \\ 0 & \frac{12EI}{\ell^3} & \frac{6EI}{\ell^2} & 0 & & \\ 0 & \frac{6EI}{\ell^2} & \frac{4EI}{\ell} & 0 & & \\ -\frac{EA}{\ell} & 0 & 0 & \frac{EA}{\ell} & 0 & 0 \\ 0 & -\frac{12EI}{\ell^3} & -\frac{6EI}{\ell^2} & 0 & & \\ 0 & \frac{6EI}{\ell^2} & \frac{2EI}{\ell} & 0 & & \end{bmatrix} \begin{bmatrix} u_1 \\ v_1 \\ \phi_1 \\ u_2 \\ v_2 \\ \phi_2 \end{bmatrix}$$

Axial deformation (u_1, u_2) & bending are decoupled.

Method of Virtual Work for Truss Elements

$$\delta W_{ext} = \delta W_{int}$$

$$\delta W_{int} =$$

$$\varepsilon = \frac{u_2 - u_1}{\ell} =$$

$$\underline{B} =$$

Strain-displacement matrix

$$\delta\varepsilon = \frac{\delta u_2 - \delta u_1}{\ell} =$$

$$\underline{\varepsilon} = \underline{B}\,\underline{u}$$

$$\delta\underline{\varepsilon} = \underline{B}\,\delta\underline{u}$$

$$\delta W_{int} =$$

Method of Virtual Work for Beam Elements

Step 1. Define strain as a function of deflection (specific to beams)

$$\underline{\varepsilon} = f(v)$$

Step 2. Define deflection as a function of nodal degrees of freedom

$$v = f(\underline{u})$$

Step 3. Derive strain-displacement matrix $\underline{B}$

$$\underline{\varepsilon} = f(\underline{u}) \rightarrow \underline{\varepsilon} = \underline{B}\,\underline{u}$$

Step 4. Use method of Virtual Work to determine $\underline{k}$ (all elements)

$$\int_V \underline{B}^T \underline{E}\, \underline{B}\, \underline{u}\, dV = \underline{f}$$

$\underline{B}$ terms are not constant in finite elements

Strain in curved beam depends on distance from neutral axis

$$\varepsilon = y\kappa \qquad \frac{d^2v}{dx^2} = \phi''$$

$$\varepsilon = y\frac{d^2v}{dx^2} \quad (\varepsilon = f(v))$$

